



Request for Proposal (RFP)

Project Title:

Convert .NET Modular Monolith (eShop) to Python with FastAPI

Engagement Type:

Project-Based Outsourcing Engagement

This is a fixed-scope, milestone-driven **project outsourcing engagement**. The selected freelancer or team will be responsible for delivering a fully functional, production-ready Python backend that mirrors the referenced .NET Modular Monolith, following strict architectural fidelity.

Project Overview:

We are seeking a qualified Python backend developer (or team) to **convert a .NET-based Modular Monolith backend** into a **Python-based FastAPI project**, while **faithfully mirroring the architecture, structure, patterns, and behavior** of the original.

- **Original Codebase:**

- [GitHub Repository](#)

- **Reference Course:**

- [Udemy Course](#)

Objectives:

- Replicate the .NET monolith architecture in Python with FastAPI
 - Preserve 1:1 structure: project layers, class/method names, DTOs, patterns
 - Implement DDD, CQRS, Outbox, Mediator, and DI in Python
 - Ensure production readiness with Docker, CI/CD, test coverage, and clean code
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Deployment Environment:

- The application must be fully **deployable and runnable on a local PC**
- Docker-based local deployment preferred (Windows or macOS support)
- No cloud deployment required

Scope of Work:

- Full code migration: API, Application, Domain, Infrastructure layers
 - Apply patterns: DDD, CQRS, Outbox, Mediator, DI, object mapping, logging
 - Use: FastAPI, SQLAlchemy, Alembic, Redis, RabbitMQ, Pydantic, structlog, py-mapper
 - API tested via Postman (Swagger disabled)
 - Min. 80% test coverage (pytest, pytest-cov)
 - Deliver fully containerized solution with CI/CD via GitHub Actions
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Tech Stack & Tooling:

- **Languages & Frameworks:** Python 3.11+, FastAPI
 - **ORM & DB:** SQLAlchemy, PostgreSQL, Alembic
 - **Messaging:** Redis, RabbitMQ
 - **Logging:** structlog → SEQ
 - **Object Mapping:** py-mapper
 - **DI:** python-dependency-injector
 - **Testing:** pytest, pytest-cov
 - **Tooling:** black, ruff, pyright, bandit, pip-audit, safety, poetry, pre-commit
 - **CI/CD:** GitHub Actions
 - **Deployment:** Docker Desktop for local environment
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Deliverables (Per Milestone):

Each milestone must include: -  Functional code (GitHub or ZIP) -  Docker setup + `.env.example` -  README & setup instructions -  Postman collection -  Coverage report ($\geq 80\%$) -  Any limitations/issues clearly documented -  Code walkthrough (2 sessions/milestone) -  Local deployment support -  Client must be able to run and validate code on their local PC

Requested Response from Freelancer:

Please include the following in your proposal:

☐ 1. Milestone Breakdown & Timeline

- Define deliverables per module aligned with the Udemy course
- Provide estimated timelines for each milestone



2. Total Cost & Payment Plan

- Overall project cost

- Proposed milestone-wise payment schedule

3. Library Inventory

- Complete list of Python packages and third-party tools you intend to use
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Additional Notes:

- All deliverables are client-owned
 - Reference architecture slides (`mma+slides.pdf`) will be shared
 - This is a **project-based outsourcing engagement** with clear acceptance criteria at each milestone
 - **All architecture decisions, design patterns, and deviations from the original must be fully documented and mutually approved in advance before implementation**
 - Limitations must be documented upfront and mutually agreed upon
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We look forward to your detailed proposal.

Best regards,

[Your Name]

[Your Organization or Team]

[Contact Email or Platform]